

UML Diagram 1: Core Domain Model

Purpose: Central entities for capacity reservation management, placement, and quota governance.

Status: Design-first (70% complete) — entity outlines with known properties; some property types and constraints TBD.

```

classDiagram
%% ===== AZURE RESOURCE LAYER =====
class CapacityReservationGroup {
    +String subscriptionId
    +String resourceGroupName
    +String name
    +String location
    +String environment
    +String crgScope
    +List~String~ zones
    +Int totalCapacityVCPUs
    +Int allocatedVCPUs
    +String sharingProfile
    +List~String~ consumerSubscriptions
    +DateTime createdAt
    +DateTime lastModified
    +getTotalCapacity() Int
    +getAllocatedVMs() Int
    +getEffectiveFree() Int
    +isZonal() Boolean
}
    note for CapacityReservationGroup "CAP-023: per environment/region the engine holds ONE regional CRG (crgScope=reg, non-zonal SKUs) PLUS one per-AZ CRG per zone (crgScope=az1|az2|az3). Environments never mixed in a CRG (ENV-003). Name follows OPS-006/C-12: crg-<env>-<region>-<scope>."

class CapacityReservation {
    +String crgId
    +String name
    +String skuName
    +Int quantity
    +String reservationType
    +String placementType
    +List~String~ zones
    +Int allocatedVMCount
    +ProvisioningState state
    +DateTime createdAt
    +getReservedVCPUs() Int
    +getAllocatedVCPUs() Int
    +isFullyAllocated() Boolean
    +isSeed() Boolean
}
    note for CapacityReservation "CAP-022: reservationType=seed means a count-0 (quantity=0) seed reservation in the seed matrix, scaled up from 0 on first allocated demand. CAP-020: placementType is Zonal (preferred) or Regional (SKU without zonal support); Availability-Set VMs are ineligible and never reserved. A 'seed reservation' (count-0) is distinct from the placement 'seed record' (CustomerSeedRecord, PLC-003)."
```

```

class SeedMatrixEntry {
    +String entryId [PK]
    +String skuFamily
    +String region
    +String availabilityZone
    +String crgScope
    +Boolean eligible
    +String owningProductTeam
    +String approvedBudgetLine
    +String scopeFileVersion
    +String governanceStatus
    +isBudgetApproved() Boolean
}

```

note for SeedMatrixEntry "CAP-022: one row per eligible SKU x region x AZ combination in the scope file (CAP-019, versioned). A combination enters the matrix only with a named owning product team AND an approved budget line. Reactive discovery (CAP-024) inserts an entry with governanceStatus=PENDING_RATIFICATION."

```
class ReservationEligibility {
    +String vmId [PK]
    +String skuName
    +String placementKind
    +Boolean reservationEligible
    +String remediationAction
    +String onboardingState
    +requiresRedeployToAZ() Boolean
}
```

note for ReservationEligibility "CAP-020/CAP-021: placementKind in {AvailabilityZone, Regional, AvailabilitySet}. AvailabilitySet VMs are reservationEligible=false and excluded from allocated/associated counts; remediationAction records the deallocate/redeploy-to-AZ onboarding precondition. The engine never auto-migrates running workloads."

```
%% ===== ACRME CONTROL PLANE ENTITIES =====
class CustomerRegionAssignment {
    +String assignmentId [PK]
    +String customerId
    +String environment
    +Region prodRegion
    +Region nonProdRegion
    +Region drRegion
    +PlacementScoreBreakdown scoreBreakdown
    +String snapshotVersionProd
    +String snapshotVersionNonProd
    +String snapshotVersionDR
    +String policyVersionId
    +List~String~ holdIds
    +AssignmentStatus status
    +DateTime assignedAt
    +DateTime validUntil
    +validateRegionDistinctness() Boolean
    +getTargetCRG(environment) CapacityReservationGroup
}
```

```
class PlacementPolicy {
    +String policyId [PK]
    +String version
    +String customerId
    +Map~String,Float~ weights
    +List~HardConstraint~ hardConstraints
    +Float drRatioMin
    +Float drRatioMax
    +Float drAutoIncreaseThreshold
    +Float prodAutoIncreaseThreshold
    +Float nonProdAutoIncreaseThreshold
    +Boolean allowNonProdDRCoLocation
    +DateTime effectiveFrom
    +DateTime supersededAt
    +evaluateHardConstraints(region) Boolean
    +calculatePlacementScore(snapshot) Float
}
```

```
class RegionalSnapshot {
    +String snapshotId [PK]
    +String region
    +DateTime capturedAt
}
```

```

+Int ttlSeconds
+Map~String,QuotaData~ quotaByFamily
+List~CRGSummary~ activeCRGs
+Int totalReservedVCPUs
+Int totalAllocatedVCPUs
+Map~String,ZoneCapacity~ zoneCapacity
+Float distanceFromProdRegion
+isStale() Boolean
+getEffectiveFreeCapacity() Int
}

class QuotaGroup {
+String groupId [PK]
+String subscriptionId
+String region
+QuotaGroupType type
+Int totalQuotaVCPUs
+Int reservedVCPUs
+Int drFloorVCPUs
+Int emergencyHeadroomVCPUs
+Int usedVCPUs
+DateTime createdAt
+getAvailableForAllocation() Int
+canAccommodate(demandVCPUs) Boolean
+enforceFloor(drDemandVCPUs) Boolean
}

class CapacityIncreaseRequest {
+String requestId [PK]
+String crgId
+RequestType type
+Int currentQuantity
+Int targetQuantity
+String reason
+IncreaseRequestStatus status
+String requestedBy
+String approvedBy
+DateTime requestedAt
+DateTime approvedAt
+DateTime completedAt
+String operationRecordId
+requiresApproval() Boolean
+canAutoApprove() Boolean
}

class EmergencyCapacityTransfer {
+String transferId [PK]
+TransferTier tier
+String sourceRegion
+String targetRegion
+String sourceCRGId
+String targetCRGId
+Int transferVCPUs
+List~String~ affectedVMIds
+TransferStatus status
+String requestedBy
+String approvedBy
+DateTime initiatedAt
+DateTime completedAt
+String operationRecordId
+requiresDualApproval() Boolean
+isQuotaNeutral() Boolean
}

```

```

class SharingRelationship {
    +String relationshipId [PK]
    +String providerCRGId
    +String consumerSubscriptionId
    +String consumerTenantId
    +SharingScope scope
    +DateTime sharedAt
    +DateTime revokedAt
    +Boolean isActive
    +getActiveVMCount() Int
}

class ZoneMappingRecord {
    +String recordId [PK]
    +String providerSubscriptionId
    +String consumerSubscriptionId
    +String region
    +Map~String,String~ zoneMapping
    +DateTime validatedAt
    +translateZone(logicalZone) String
    +isMappingAvailable() Boolean
}

%% ===== ENUMERATIONS =====
class QuotaGroupType {
    <<enumeration>>
    PROD
    NONPROD_DR_SHARED
}

class AssignmentStatus {
    <<enumeration>>
    PENDING
    ACTIVE
    HOLD
    EXPIRED
    REVOKED
}

class IncreaseRequestStatus {
    <<enumeration>>
    PENDING_APPROVAL
    APPROVED
    IN_PROGRESS
    COMPLETED
    FAILED
    CANCELLED
}

class TransferStatus {
    <<enumeration>>
    INITIATED
    QUOTA_VALIDATED
    REDUCING_SOURCE
    EXPANDING_TARGET
    COMPLETED
    FAILED
    ROLLED_BACK
}

class TransferTier {
    <<enumeration>>

```

```

TIER_1_DIRECT_EXPANSION
TIER_2_QUOTA_NEUTRAL_TRANSFER
TIER_3_DESTRUCTIVE_TRANSFER
}

%% ===== RELATIONSHIPS =====
CapacityReservationGroup "1" --o "*" CapacityReservation : contains
CapacityReservationGroup "1" o-- "*" SharingRelationship : provider
CustomerRegionAssignment "*" --> "3" CapacityReservationGroup : targets (Prod|Non-
Prod|DR)
CustomerRegionAssignment "*" --> "1" PlacementPolicy : evaluatedBy
CustomerRegionAssignment "*" --> "3" RegionalSnapshot : usesSnapshots
QuotaGroup "1" --> "*" CapacityReservationGroup : governs
CapacityIncreaseRequest "*" --> "1" CapacityReservationGroup : targets
EmergencyCapacityTransfer "*" --> "1" CapacityReservationGroup : source
EmergencyCapacityTransfer "*" --> "1" CapacityReservationGroup : target
SharingRelationship "*" --> "1" ZoneMappingRecord : requiresZoneMapping
PlacementPolicy --> QuotaGroupType : references
QuotaGroup --> QuotaGroupType : type
CustomerRegionAssignment --> AssignmentStatus : status
CapacityIncreaseRequest --> IncreaseRequestStatus : status
EmergencyCapacityTransfer --> TransferStatus : status
EmergencyCapacityTransfer --> TransferTier : tier
SeedMatrixEntry "1" --> "0..1" CapacityReservation : seeds (count-0)
SeedMatrixEntry "*" --> "1" CapacityReservationGroup : placedIn (per-AZ or region-
al)
ReservationEligibility "*" --> "0..1" CapacityReservation : eligibleFor
CapacityReservationGroup --> ReservationType : holds
class ReservationType {
    <<enumeration>>
    SEED
    ACTIVE
}
class CRGScope {
    <<enumeration>>
    REGIONAL
    AZ1
    AZ2
    AZ3
}
class PlacementKind {
    <<enumeration>>
    AVAILABILITY_ZONE
    REGIONAL
    AVAILABILITY_SET
}
class GovernanceStatus {
    <<enumeration>>
    RATIFIED
    PENDING_RATIFICATION
    REJECTED
}

```

v2.4 additions to the core domain model. Three concepts from Baseline v2.4 are now modelled: (1) `SeedMatrixEntry` + `CapacityReservation.reservationType=SEED` capture the **seed-at-0 matrix and budget governance** (CAP-022); (2) `CapacityReservationGroup.crgScope` (`REGIONAL / AZ1..AZ3`) + `environment` capture the **regional + per-AZ CRG structure** (CAP-023) with OPS-006/C-12 naming; (3) `ReservationEligibility` + `CapacityReservation.placementType` capture **Availability-Set ineligibility and the deallocate/redeploy-to-AZ onboarding precondition** (CAP-020/CAP-021). Reactive discovery (CAP-024) is a service-layer flow (uml_03) that inserts a `SeedMatrixEntry` with `governanceStatus=PENDING_RATIFICATION` and an `ACTIVE` reservation sized `allocated + buffer`.

Known Gaps (to be resolved in design session):

Entity Schemas (incomplete properties):

- **CustomerRegionAssignment:** Missing `PlacementScoreBreakdown` structure (alpha, beta, gamma, delta, epsilon values)
- **RegionalSnapshot:** Missing exact `CRGSummary`, `QuotaData`, `ZoneCapacity` nested structures
- **PlacementPolicy:** Missing `HardConstraint` definition (HC-1 through HC-7 structure)
- **QuotaGroup:** Exact Azure Quota API semantics for `totalQuotaVCPU`s vs `usedVCPU`s TBD
- **All entities:** Cosmos DB partition keys, indexes, TTL settings TBD

Relationships (cardinality/constraints TBD):

- `CustomerRegionAssignment` → `CRG`: geography-aware region-distinctness constraint — three-region geography (US) requires Prod, CVAL, DR distinct; two-region geography (EU/Australia/Asia Pacific) requires Prod distinct with **CVAL + DR co-located** (PLC-010a); **[Amended v2.4] the Middle East is a cross-geo DR geography** (PLC-010b, DR-020, DEC-001 RESOLVED) — Prod + CVAL co-located in a weighted-selected Middle East region (separate CRGs) with **DR placed cross-geo in a weighted-selected Europe region** (enforced in placement validation / PlacementPolicy)
- `QuotaGroup` → `CRG`: One group can contain multiple CRGs in same region, exact cardinality TBD
- `SharingRelationship`: 100-consumer hard limit per CRG (enforced in entity or service layer?)

Business Rules (not yet modeled):

- DR floor enforcement: `drFloorVCPU`s ≤ allocated DR capacity (where enforced?)
- Emergency headroom staging: How is `emergencyHeadroomVCPU`s pre-allocated in `QuotaGroup`?
- Auto-increase trigger thresholds: Separate per environment (0.35 DR, 0.20 Prod/NonProd) — stored where?